

AI-Powered Customer Support Chatbot Platform with Multi-Channel Integration



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Declaration

I, Ruchi Baweja, hereby declare that the project report titled "**AI-Powered Customer Support Chatbot Platform with Multi-Channel Integration**" submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (MCA) is my original work carried out under the guidance of Vasanthi Chandran. The work presented in this report has not been submitted previously for any other degree or certification.

Acknowledgement

I express my sincere gratitude to my mentor, faculty members, and institution for their continuous support, guidance, and encouragement throughout the development of this project. Their valuable suggestions helped in successful completion of this work.

Abstract

This project focuses on the design and development of an AI-powered customer support chatbot platform integrated across multiple communication channels. The system is capable of handling user queries using Natural Language Processing (NLP), automated response generation, and live agent escalation. The platform supports web and mobile communication channels and aims to improve customer engagement and service efficiency. The project integrates chatbot architecture, intent recognition, entity extraction, response databases, analytics dashboards, and secure conversation history management. Technologies such as React.js, Node.js, MongoDB, Python, and NLP frameworks are utilized to develop the platform. The system also provides scalability, security, and efficient customer support automation for modern organizations.

Chapter 1: Introduction

Customer support automation has become essential for modern businesses. Traditional support systems require human agents for handling repetitive customer queries, which increases operational cost and response time. AI-powered chatbot systems help automate customer interaction using NLP and machine learning technologies. The proposed system provides an intelligent chatbot capable of understanding user queries, generating automated responses, and escalating complex issues to live support agents. The system improves efficiency, customer satisfaction, and scalability.

Project Objectives

- Design and develop an AI-powered customer support chatbot platform.
- Handle user queries across web and mobile channels.
- Integrate NLP-based response generation.
- Implement live agent escalation features.
- Improve customer engagement and support efficiency.

Project Tasks

- Study chatbot architecture and NLP fundamentals.
- Design workflow for user query processing.
- Integrate NLP libraries such as Dialogflow or Rasa.
- Develop intent recognition and entity extraction modules.
- Create predefined response databases.
- Implement machine learning-based response improvement.
- Develop multi-channel integration.
- Add live agent handover feature.
- Store conversation history securely.
- Implement analytics dashboard for query trends.
- Conduct testing for accuracy measurement.
- Optimize response time performance.

Chapter 2: System Analysis

The system consists of multiple modules including chatbot engine, analytics dashboard, admin panel, authentication system, and customer interaction channels. Functional requirements include automated responses, secure login, conversation storage, and real-time communication. Non-functional requirements include scalability, reliability, security, and performance optimization.

Chapter 3: System Design

The system architecture follows a client-server model. The frontend interface communicates with backend APIs. NLP processing is performed using AI libraries. MongoDB stores user conversations and analytics data. UML diagrams, use case diagrams, and ER diagrams are used to represent the overall structure of the system.

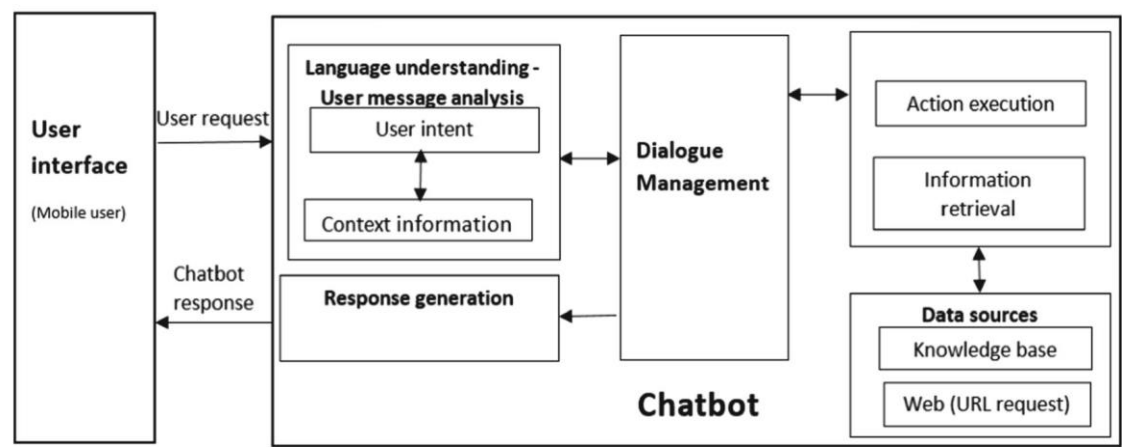


Figure 3.1: System Architecture of AI-Powered Customer Support Chatbot Platform

Explanation : The above architecture diagram represents the overall workflow of the chatbot platform. User queries are processed through the frontend interface and transferred to the backend server where NLP processing and response generation are performed. The database stores user conversations and analytics information.

Figure 3.2 — Use Case Diagram

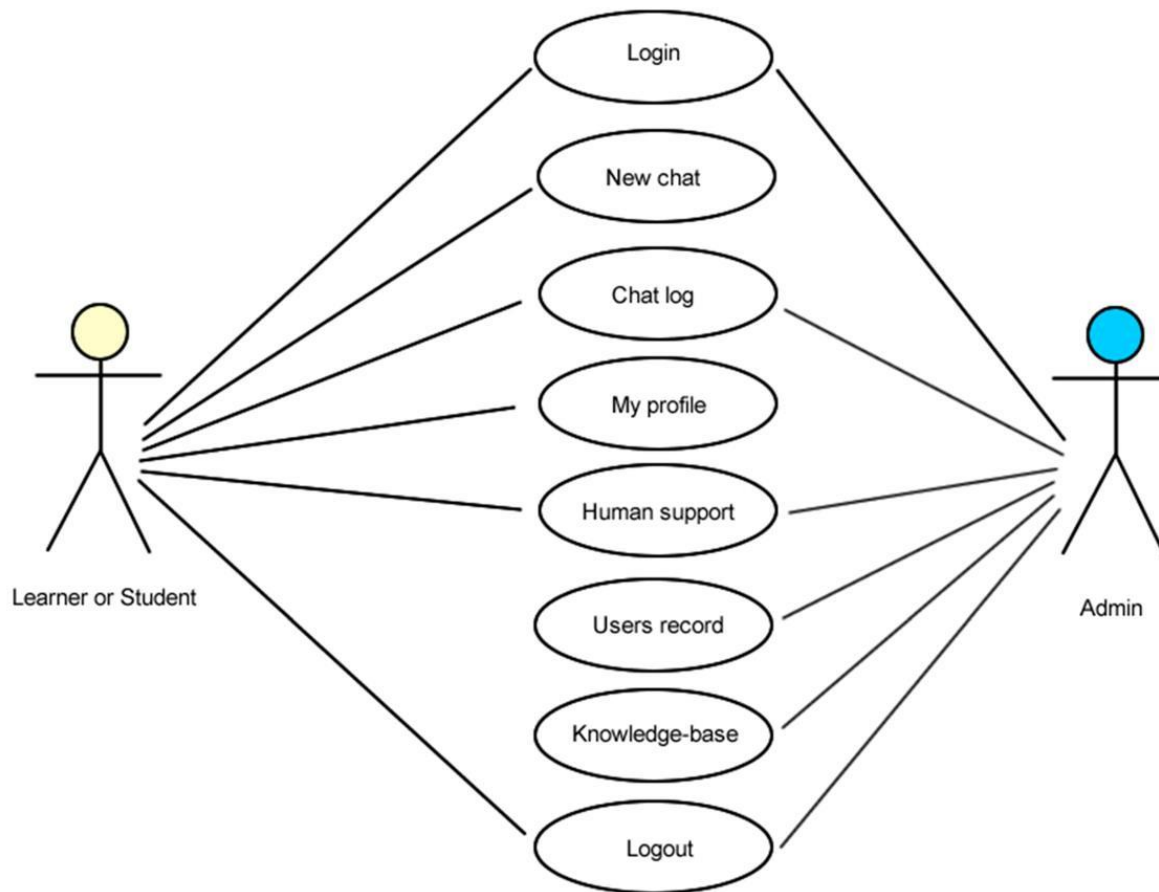


Figure 3.2: Use Case Diagram of Customer Support Chatbot Platform

Explanation : The use case diagram illustrates interactions between customers, administrators, and support agents with the system. Customers can send queries and receive responses, while administrators manage chatbot operations and analytics.

Chapter 4: Technology Stack

Frontend: React.js

Backend: Node.js & Express.js

Database: MongoDB

AI/NLP: Python, Dialogflow/Rasa

Real-Time Communication: Socket.io

Version Control: Git & GitHub

Chapter 5: Testing

The system undergoes unit testing, integration testing, and system testing. Various test cases are executed to validate chatbot responses, API integration, and real-time communication. Performance testing ensures fast response times and stable operation under multiple user requests.

Chapter 6: Conclusion & Future Scope

The proposed AI-powered customer support chatbot platform successfully automates customer interaction and provides efficient support services. The system improves communication speed and reduces manual workload. Future enhancements include multilingual support, voice assistant integration, cloud deployment, CRM integration, and advanced AI-based analytics.